SentinelAI

GitGuardian tells you your house is on fire.

SentinelAI puts it out.

Zinnia Nagpal · cmd-f 2026

// the problem

Secrets leak into code every day.

"I pushed an API key to GitHub. It was compromised"

3 min	85%	$0
avg time to exploit a leaked key	of devs have committed secrets	cost of prevention vs. breach

Existing tools detect leaks AFTER the push. Too late.

// the solution

SentinelAI

A pre-push git hook that detects, replaces, and vaults secrets automatically.

git push
dev tries to push

→

hook fires
pre-push intercepts

→

scan diff
regex + entropy

→

vault it
encrypt + store

→

push blocked
secret replaced

```
Before: OPENAI_API_KEY = "sk-abc123..."     →     After: OPENAI_API_KEY =
"op://vault/sentinelai/openai-key"
```

// how it detects secrets

Two layers of detection

🎯 Regex Pattern Matching

- → AWS Access Keys
- → OpenAI / Anthropic Keys
- → GitHub / GitLab Tokens
- → Stripe / Twilio Keys
- → JWT Tokens
- → Private Keys (RSA, EC)
- → 30+ patterns total

📊 Shannon Entropy

Catches secrets that don't match any known pattern — by measuring how randomly distributed characters are.

$$H = -\Sigma\ p(x) \cdot \log_2(p(x))$$

Normal text: 2.5 entropy
Real secrets: **3.5–4.5+ entropy**

// features

A complete security pipeline

🔍 Smart Detection

30+ regex patterns + Shannon entropy catches both known and unknown secrets

🔄 Auto-Remediation

Replaces raw secrets with op:// vault references directly in your files

🔒 Encrypted Vault

AES Fernet encryption stores secrets locally nothing sent to the cloud

📲 Slack Alerts

Real-time team notifications the moment a push is blocked

📋 Audit Log

Full history of every push scanned who, when, what secrets, what action

🛡 Dashboard

Live Streamlit dashboard with terminal simulation and vault browser

// live demo

Watch it happen in real time

bash — demo-repo

```
$ git push origin main
Enumerating objects: 3, done.
⚡ SentinelAI pre-push hook triggered
  Scanning staged diff for secrets...
  🚨 SECRETS DETECTED IN DIFF
  ⚠️ OpenAI Key → replaced with op://vault/sentinelai/openai-key
  ⚠️ AWS Access Key → replaced with op://vault/sentinelai/aws-access-key
  🔒 Vaulted securely | 📲 Slack alert sent
❌ PUSH REJECTED — review changes and re-push
```

// built with · what's next

Built with

- Python
- SQLite
- Fernet AES Encryption
- Streamlit
- Slack Webhooks
- Shannon Entropy
- Regex (30+ patterns)

What's next

→ 1Password SDK — cloud vault integration

→ sentinelai install — one command setup

→ VS Code extension — catch as you type

→ Screen monitoring — catch during screen share

→ ML classifier for unknown secret types

→ Team dashboard across multiple repos

SentinelAI

Don't just detect the fire.

Put it out.

github.com/Zinnia-Nagpal/ai-security-battle

Built solo at cmd-f 2026 · Zinnia Nagpal